~~So~~

Udemy ML, DL, NLP course

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Sec - 29

Logistic Regression

Lec 1: Can Linear Regression solve classifier Problem.

Lec 2: Logistic Regression Indepth Math Intuition

Logistic Regression

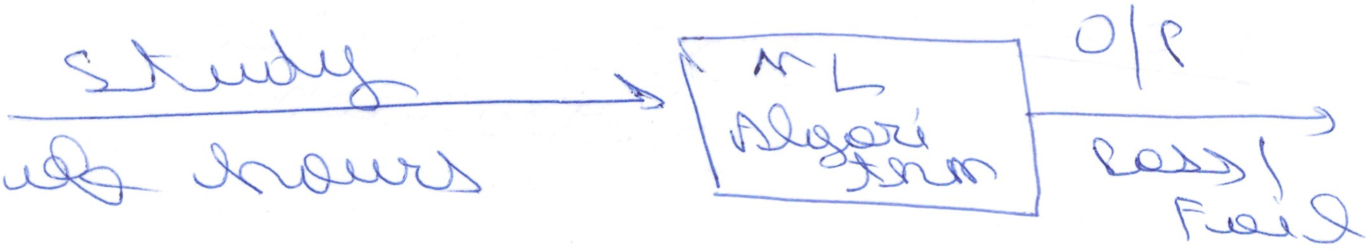
(Binary Classification)

Dataset

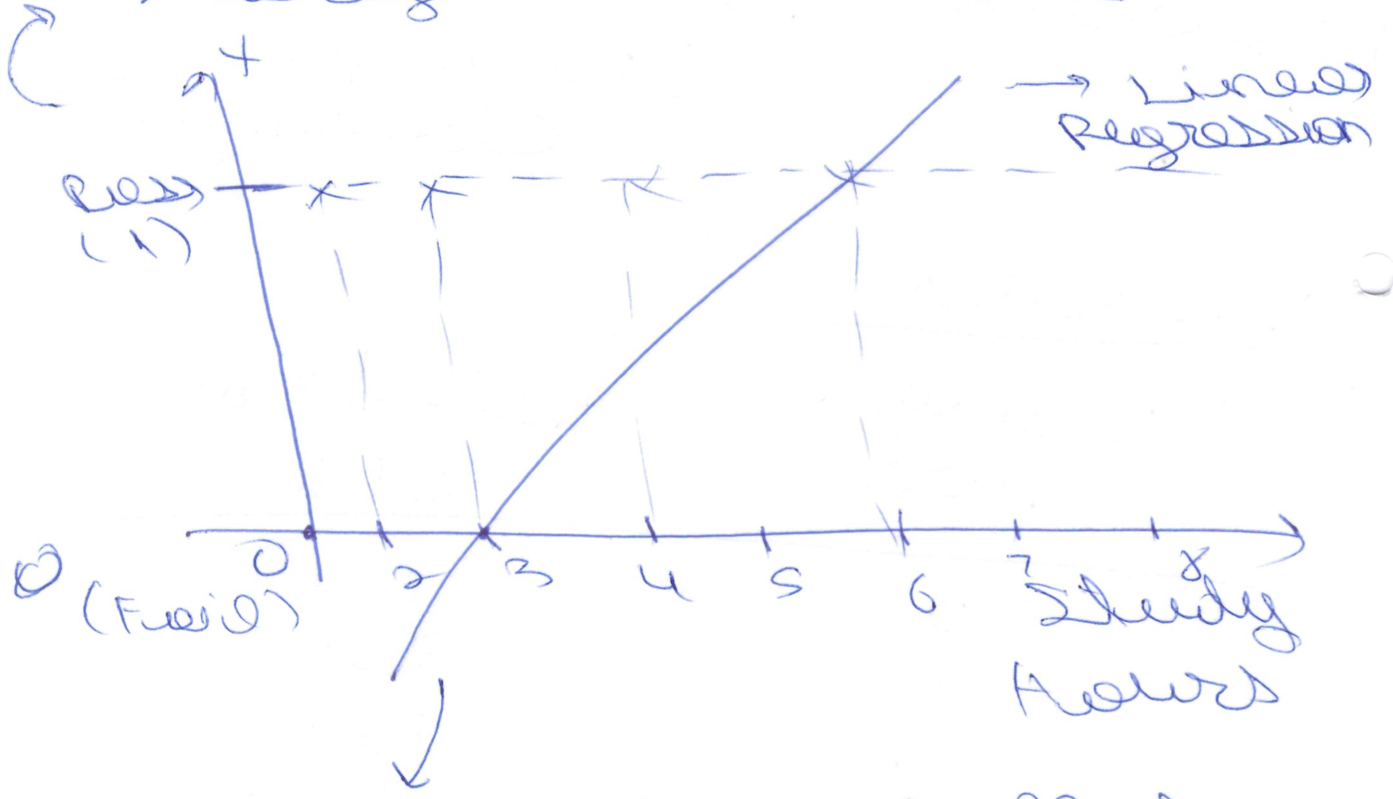
Study Hours	Pass/Fail O/P
2	Fail
3	Fail
4	Fail
5	Pass
6	Pass
7	Pass

In the dataset, pass/ fail depending on no. of hours studied.

We want a ML algorithm which takes study of hours as output & gives



2 categories as + Pass



minimal Error will try to create best fit line

We treated this with help of Linear regression.

If value $\leq 0.5 \Rightarrow 0$

& if value $\geq 0.5 \Rightarrow 1$

→ So this is how we try solving by Linear regression.

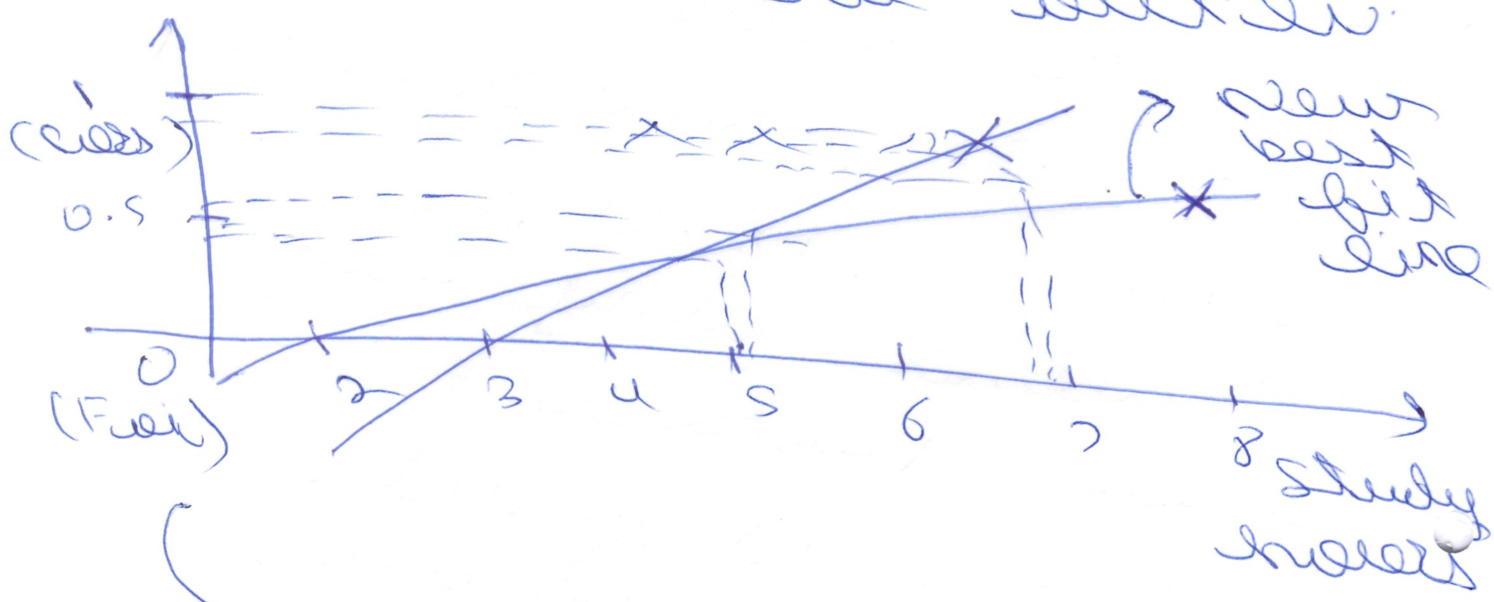
→ Here mid point is taken as 0.5.

Hence with Linear regression also, we're able to solve it for any new data points.

⇒ Problem in above method.

Now suppose we add an outlier of 12 hours

Here we have an error to make the best fit line inclined to outlier.



→ Because of an outlier, entire line is getting changed.

→ We also get values greater than 1, & less than 0

→ Whereas this is not fit for binary classification.

⇒ Why we cannot use linear regression for classification?

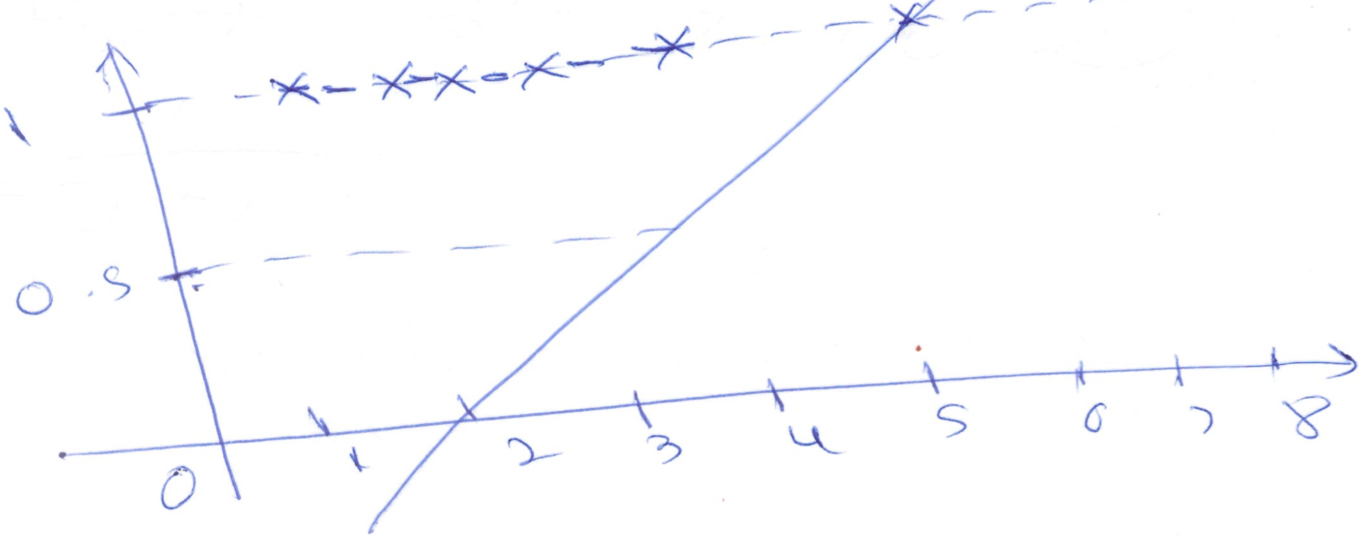
1. Outlier (Best fit line change)

2. output > 1 & < 0

↳ It leads the squashing that line.

Logistic Regression Math Intuition

⇒ How Logistic regression solves classification problem



$$\boxed{h_\theta(x) = \theta_0 + \theta_1 x_1}$$

↳ This is of Linear Regression & gives us the best fit line

=> ~~How~~ ↳ But here we don't want this specific equation.

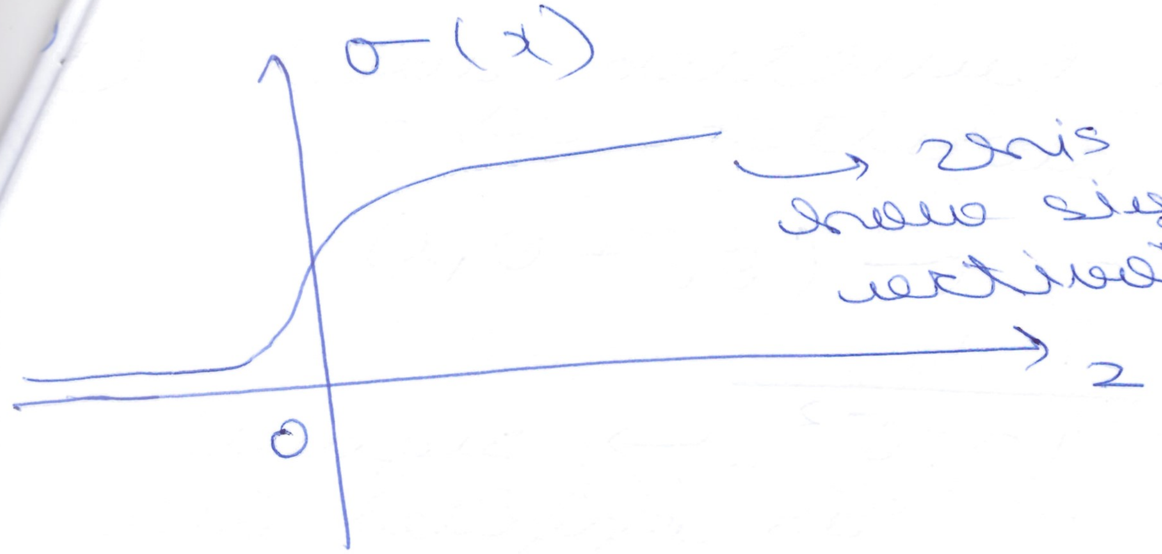
Because, from here, we really need to squash it

=> And how squashing is basically possible, that we have to see.

=> Sigmoid Activation

↳ Given by $\frac{1}{\cancel{1+e^{-x}}}$

$$\frac{1}{1+e^{-z}}$$



→ This is sigmoid curve activation function will be

sigmoid ranges outputs b/w 0 to 1

→ So basically it means that value will always be from 0 to 1.

whenever $z > 0$

→ $\sigma(z) > 0.5$

→ sigmoid of z greater than or equal to 0.5

→ we apply sigmoid activation on best fit line.

⇒ New Function will be written as :-

$$h_\theta(x) = \sigma(\theta_0 + \theta_1 x)$$

$$\sigma = \frac{1}{1+e^{-z}}$$ → sigmoid is applied like this.

$$\sigma = \frac{1}{1+e^{-z}} \qquad z = \theta_0 + \theta_1 x_1$$

$$h_\theta(x) = \sigma(\theta_0 + \theta_1 x)$$
$$= \sigma(z)$$

$$h_\theta(x) = \frac{1}{1+e^{-z}}$$

$$z = \theta_0 + \theta_1 x_1$$

→ z is basically this value.

$$h_\theta(x) = \frac{1}{1+e^{-z}}$$

$$z = \theta_0 + \theta_1 x_1$$

Logistic Regression Hypothesis.

Linear Regression cost Function

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x)^{(i)} - y^{(i)}\right)^2$$

$$h_\theta(x) = \theta_0 + \theta_1 x$$

→ It gives convex function

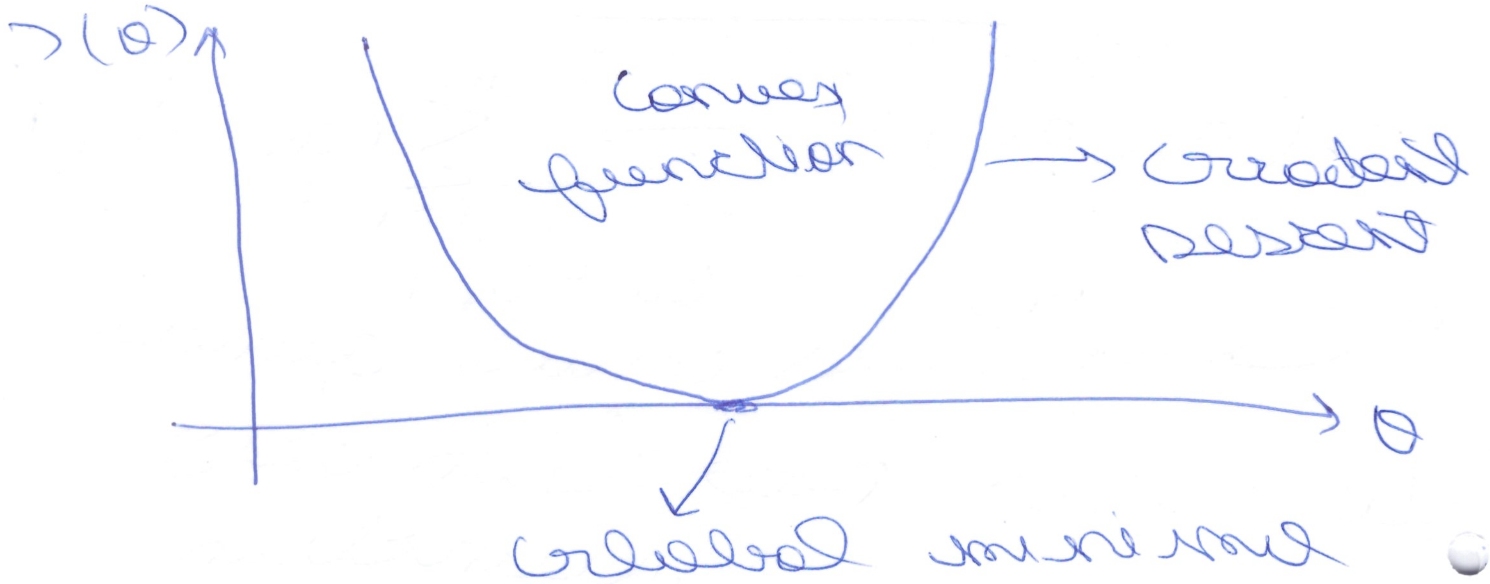
Logistic Regression cost function

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x)^{(i)} - y^{(i)}\right)^2$$

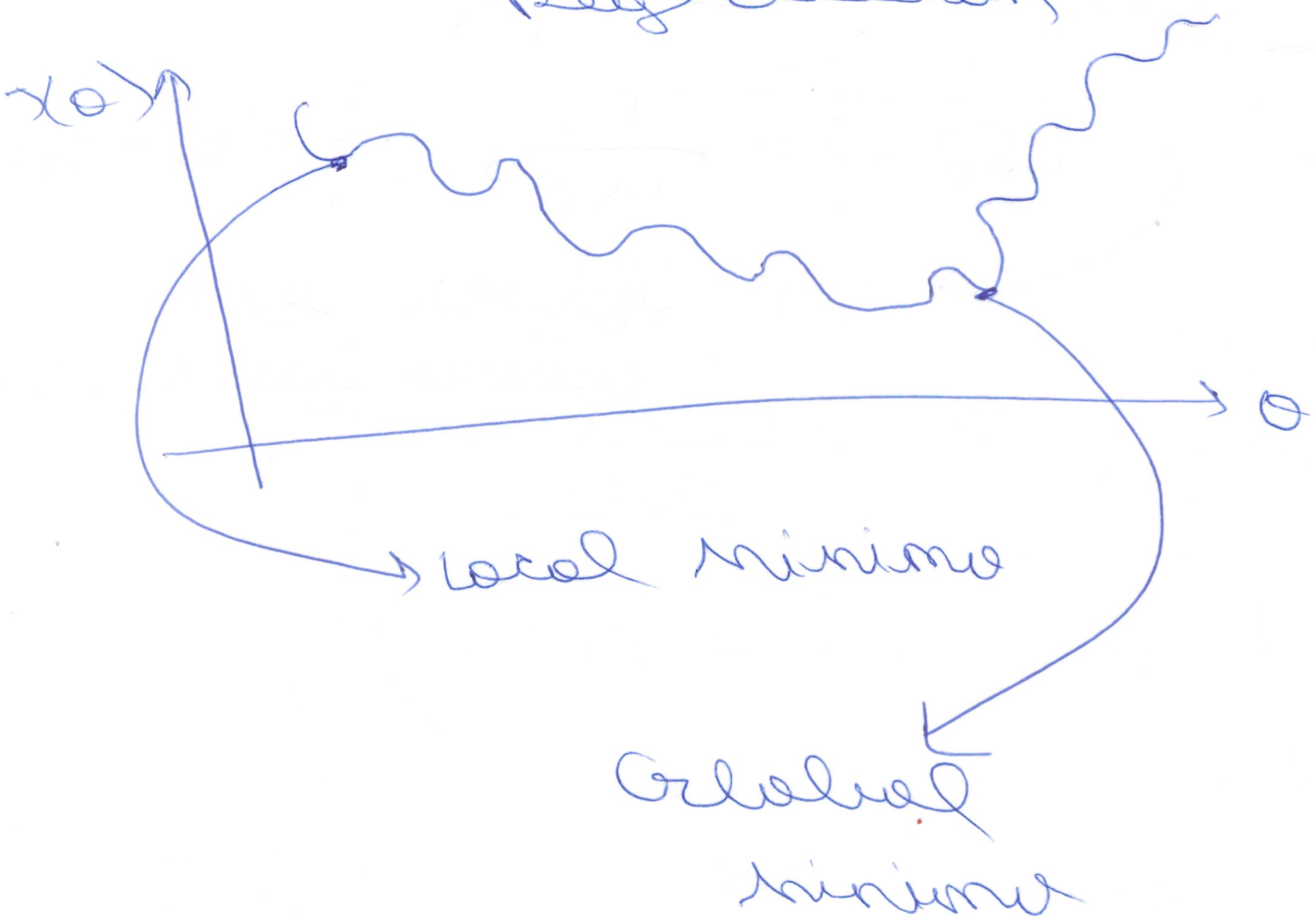
$$h_\theta(x) = \frac{1}{1+e^{-z}} \quad z = \theta_0 + \theta_1 x$$

→ It gives non convex function

=> Curve for linear regression



=> Curve for Logistic regression



$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x)^i - y^{(i)} \right)^2$$

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$$h_\theta(x)^{(i)} = \frac{1}{1+e^{-z}} \quad z = \theta_0 + \theta_1 x_1$$

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \underbrace{\left(h_\theta(x)^{(i)} - y^{(i)} \right)^2}$$

$$\mathrm{cost}\left(h_\theta(x)^{(i)}, y^{(i)}\right)$$

Let's denote $\mathrm{cost}\left(h_\theta(x)^i, y^{(i)}\right)$

$$\mathrm{cost}\left(h_\theta(x)^i, y^{(i)}\right) = \begin{cases} -\log(h_\theta(x)) & \text{if } y = 1 \\ -\log(1 - h_\theta(x)) & \text{if } y = 0 \end{cases}$$

$$\text{Cost}(h_\theta(x)^i, y^{(i)})$$

$$= -y \log(h_\theta(x)^i) - (1-y)(1-h_\theta(x))$$

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} \left(y^{(i)} \log(h_\theta(x)^{(i)}) - (1-y^{(i)}) \log(1-h_\theta(x)^{(i)}) \right)$$

→ minimize cost function $J(\theta_0, \theta_1)$ by changing θ_0 & θ_1